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StoryCube: Tangible Prompt Engineering for Pre-Literate Children

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ABSTRACT

Preschool children are largely excluded from interacting with LLMs because current prompt interfaces rely on fluent speech, writing, or reading. As generative AI increasingly pervades children's everyday environments, they need age-appropriate, accessible ways to express intent and gain agency in AI-mediated experiences. We introduce StoryCube, a tangible prompt engineering interface that lets children compose structured prompts using physical story tokens to generate audio stories. StoryCube is novel in enabling structured prompting through tangible interaction, supporting pre-literate access and collaborative, hands-on prompt composition rather than text- or voice-based prompting. In a preliminary study with six children (ages 5–8) and two educators, children used StoryCube intuitively with minimal guidance, explored diverse token combinations, and reported high enjoyment, while also noting mixed story quality. Findings suggest that tangible prompt engineering is a promising concept for child-centered AI interaction, motivating further research on parametrization, token grammar, safety, and educational integration.

HOW STORYCUBE WORKS

Children select up to four NFC story tokens and place them on the cube. Each placement triggers spoken confirmation. One large red button generates, pauses, or resumes the story.

The current prototype offers 19 tokens across four color-coded categories. These choices are mapped into a structured prompt that requests a short, child-friendly story.

SUBJECT | PLOT | LOCATION | ADJECTIVE

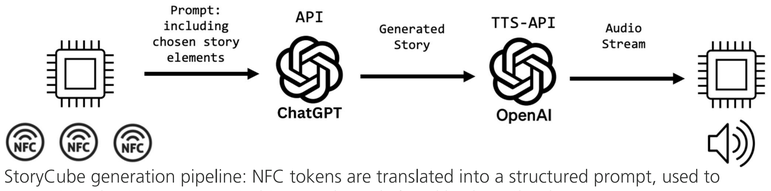
Selected tokens are read via NFC, assembled into a structured prompt, sent to the OpenAI API for story generation, converted to speech, and streamed back for audio playback.

Tangible story tokens to compose prompts.

The StoryCube device with attached tokens.

PROTOTYPE

- Standalone, USB-powered 3D-printed cube with Raspberry Pi 5, four RFID readers, speaker, volume dial, and a single generate/play/pause button.
- Magnetic token slots make placement robust and tactile for young children.
- A fixed system prompt aims for simple language, clear narrative structure, and balanced gender representation.



StoryCube generation pipeline: NFC tokens are translated into a structured prompt, used to generate a short story, converted to speech, and played back on the device.

WHY THIS MATTERS

- Most LLM interfaces assume reading, writing, or fluent speech.
- Pre-literate children often need adults to translate their ideas into prompts and to interpret the output.
- Tangible, audio-first interaction can give children more direct agency in AI-mediated storytelling.

A preschooler places tangible story tokens on StoryCube to compose a short story.

PRELIMINARY STUDY

Deployment took place in a day-care center. Children explored StoryCube in a familiar classroom environment, followed by child and educator group interviews.

6	2	3	30 min
children ages 5-8	educators	days of use	free-play sessions

Data Sources

- In-situ observations during free exploration.
- System logs of token use, story generations, stories, and playback.
- Semi-structured group interviews analyzed with open coding.

CONTRIBUTION

- A working prototype for tangible prompt engineering for pre-literate children.
- Early evidence of intuitive, collaborative use.
- A design direction for child-centered AI that raises important questions around safety, quality control, and educational use.
- *Next:* language adaptation, strong safety guardrails, richer token grammars, longer-term study.

KEY FINDINGS

Intuitive From the Start
Children successfully operated StoryCube from the start, attached tokens, triggered stories, and explored new combinations without adult assistance.

Strong Engagement and Collaborative Play
Children described StoryCube as fun, wanted to use it again, and naturally adopted turn-taking and parallel exploration around the shared device.

Story Quality Remains Design Challenge
Children liked that stories reflected their token choices, but they also wanted more dramatic plots, more varied voices, and fewer repeated character names.

Learning Potential
Educators viewed StoryCube as a promising language-learning tool, especially if stories and language can adapt to children's needs, while emphasizing that AI for young children must remain age-appropriate, guided, and carefully safeguarded.

> "it's cool that the stories are always different."
>
> "i just attach the tokens, and it works."

TAKEAWAY

Tangible prompt engineering is a promising direction for child-centered AI: it makes intent expression accessible and collaborative, while surfacing new questions around safety, output quality, and educational integration.

Children gathered around the StoryCube, negotiated token choices, and often played together.